

Juniper Networks[®] vMX Release Notes

Release 17.1R1
21 July 2020

These release notes accompany this release of the Juniper Networks[®] virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

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Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 17.1 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

New and Changed Features

This section describes the new features and enhancements in this release.

- **CentOS support**—vMX can be installed on a host running KVM on CentOS 7.2 operating system. You must prepare the host before installing vMX.
- **Class of Service support for drop profiles**—You can manage congestion using drop profiles and packet loss priority. Packet loss priority (PLP) values are low, medium-low, medium-high, and high. Each packet loss priority value is mapped to a color that determines the behavior applied to an oversubscribed queue; low PLP maps to green, medium-low PLP and medium-high PLP map to yellow, and high PLP maps to red. Drop profiles are used to set the thresholds within a queue for a given loss priority.
- **Class of Service support for L2TP LNS**—Hierarchical scheduling and per-session shaping can be applied to L2TP network server (LNS) inline service interfaces using a static or dynamic CoS configuration. You enable the inline service interface (si) for vMX by configuring the **inline-services bandwidth (1g | 10g)** option at the **[edit chassis fpc 0 pic 0]** hierarchy level. You must also create the loopback device for each si interface by configuring the **loopback-device-count device-count** option at the **[edit chassis fpc 0]** hierarchy level where *device-count* is the number of si interfaces in the range of 1 through 4; the default value is 0 and creates no loopback devices. You enable CoS by configuring the **flexible-queuing-mode** option at the **[edit chassis fpc 0]** hierarchy level.

NOTE: The FPC reboots if you enable inline service interfaces or CoS or if you change the loopback device count.

- **DPDK support**—vMX uses DPDK 2.x to support improved performance. Scheduling is performed in a separate core and the traffic load is balanced depending on the number of vCPUs for the VFP. If there are not enough cores available, scheduling shares the I/O core and impacts performance. For optimal performance, we recommend a minimum of $(2.5 * \text{number-of-ports}) + 3$ vCPUs for VFP (without CoS), a minimum of $(3.5 * \text{number-of-ports}) + 3$ vCPUs for VFP (with CoS), and 1G Huge Pages. You must set up Huge Pages when preparing the Ubuntu host.
- **Flow caching is enabled for L2TP tunnel interfaces**—Flow caching is automatically enabled for L2TP tunnel interfaces.
- **Red Hat Enterprise Linux support**—vMX can be installed on a host running KVM on Red Hat Enterprise Linux 7.2 operating system. You must prepare the host before installing vMX.
- **Red Hat OpenStack Platform support**—Red Hat OpenStack Platform 8 is supported.
- **Virtual broadband network gateway**—Using vMX, you can deploy a virtual broadband network gateway (vBNG) on x86 servers to terminate broadband subscribers. vBNG supports most of the subscriber management features available with Junos OS Release 17.1 on MX Series routers.

vBNG runs on vMX, so it has similar exceptions; the following subscriber management features available on MX Series routers are not supported for vBNG:

- High availability features such as hot-standby backup for enhanced subscriber management and MX Series Virtual Chassis.

To deploy a vBNG instance, you must purchase these licenses:

- vMX PREMIUM application package license with 1 Gbps, 5 Gbps, 10 Gbps, or 40 Gbps bandwidth
- vBNG subscriber scale license with 1000, 10 thousand, 100 thousand, or 1 million subscriber sessions for one of these tiers:

Tier	Description
Introductory	L2TP features including L2TP LNS services, secure policy, service activation and deactivation
Preferred	Features in the Introductory tier, and DHCP subscriber services, PPP/LAC subscriber services, DHCP relay and DHCP local server
Elite	Features in the Preferred tier, and pseudowire ingress termination

With the appropriate vMX PREMIUM license, you can evaluate vBNG without a vBNG subscriber scale license for 30 days. After 30 days, you are limited to 10 subscriber sessions.

[See [Junos OS Broadband Subscriber Management and Services Library](#).]

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge. PR1059231
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- Output for the **show pfe statistics traffic** command might take a few seconds to appear and many **MIB2D_COUNTER_DECREASING: pfes_stats_delta: counter** messages appear in the log file before the response time improves. PR1071659
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902
- Multiple vMX instances using SR-IOV on the same host are not supported. PR1130534

- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync_update_tod_to_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344
- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK_SCHED_SLIP** log messages. PR1153290
- Observing **rpdd[7116]: %DAEMON-6: ifl_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096
- GRE is not supported in performance mode. PR1328554

Subscriber Management

- Dual-stack (DHCPv6 over PPPoE) subscribers can be lost on an LNS after the subscriber management process is restarted on vMX. PR1165378
- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154
- After adding or deleting the l2tp-inline-lns license on vMX and rebooting vMX, fxp0.0 was not reachable. As a workaround, access vMX through the console and reboot vMX. PR1173990

Resolved Issues in This Release

- The VMXNET3 driver does not support large MTU. PR1120215

System Requirements

IN THIS SECTION

- [Minimum Hardware and Software Requirements for KVM | 6](#)
- [Minimum Hardware and Software Requirements for VMware | 9](#)

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

[Table 1 on page 6](#) lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.

Table 1: Minimum Hardware Requirements for KVM (continued)

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port. The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the formula to calculate the minimum vCPUs needed by the VFP. To calculate the optimal number of vCPUs needed by VFP for performance mode: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.	For lab simulation use case applications (lite mode): Minimum of 4 • 1 for VCP • 3 for VFP For low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications (performance mode): Minimum of 9 • 1 for VCP • 8 for VFP
Memory NOTE: Performance mode is the default mode.	For lab simulation use case applications (lite mode): Minimum of 5 GB • 1 GB for VCP • 4 GB for VFP For low-bandwidth or high-bandwidth applications (performance mode): Minimum of 16 GB • 4 GB for VCP • 12 GB for VFP Additional 2 GB recommended for host OS
Storage	Local or NAS

Table 1: Minimum Hardware Requirements for KVM (*continued*)

Description	Value
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

Table 2 on page 8 lists the software requirements for Ubuntu.

Table 2: Minimum Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 14.04.1 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages NOTE: Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev NOTE: libvirt 1.2.19

Table 3 on page 8 lists the software requirements for Red Hat Enterprise Linux.

Table 3: Minimum Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5
Virtualization	QEMU-KVM 1.5.3
Required packages NOTE: SR-IOV requires these packages: kernel-devel gcc	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools NOTE: libvirt 1.2.17 or later

[Table 4 on page 9](#) lists the software requirements for CentOS.

Table 4: Minimum Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools NOTE: libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

[Table 5 on page 10](#) lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.	For performance mode: Minimum of 8 • 1 for VCP • 7 for VFP For lite mode: Minimum of 4 • 1 for VCP • 3 for VFP
Memory NOTE: Performance mode is the default mode.	For performance mode: Minimum of 16 GB • 4 GB for VCP • 12 GB for VFP For lite mode: Minimum of 10 • 2 GB for VCP • 8 GB for VFP
Storage	Local or NAS

[Table 6 on page 10](#) lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or JNASC support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf> .
- Product warranties—For product warranty information, visit <https://www.juniper.net/support/warranty/> .
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <http://www2.juniper.net/kb/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://www.juniper.net/alerts/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Open a case online in the CSC Case Management tool: <https://www.juniper.net/cm/>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://tools.juniper.net/SerialNumberEntitlementSearch/>

Opening a Case with JTAC

You can open a case with JTAC on the Web or by telephone.

- Use the Case Management tool in the CSC at <https://www.juniper.net/cm/> .
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://www.juniper.net/support/requesting-support.html> .

Revision History

21 July 2020—Revision 4, Junos OS Release 17.1R1—vMX

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9 May 2017—Revision 2, Junos OS Release 17.1R1—vMX

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